FORM TP 2019004

CANDIDATE – PLEASE NOTE!

Print your name on the line below and return this booklet with your answer sheet. Failure to do so may result in disqualification.

TEST CODE **01212010**

JANUARY 2019

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE® EXAMINATION

CHEMISTRY

Paper 01 – General Proficiency

1 hour 15 minutes

18 JANUARY 2019 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 15 minutes to answer them.

2. In addition to this test booklet, you should have an answer sheet.

3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.

4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

   Sample Item

   The SI unit of length is the

   (A) metre
   (B) newton
   (C) second
   (D) kilogram

   Sample Answer

   ● (B) (C) (D)

   The best answer to this item is "metre", so (A) has been shaded.

5. If you want to change your answer, erase it completely before you fill in your new choice.

6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.

7. You may do any rough work in this booklet.

8. Figures are not necessarily drawn to scale.

9. You may use a silent, non-programmable calculator to answer items.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. Which of the following processes demonstrates that matter is made up of minute particles?

(A) Diffusion
(B) Capillarity
(C) Distillation
(D) Evaporation

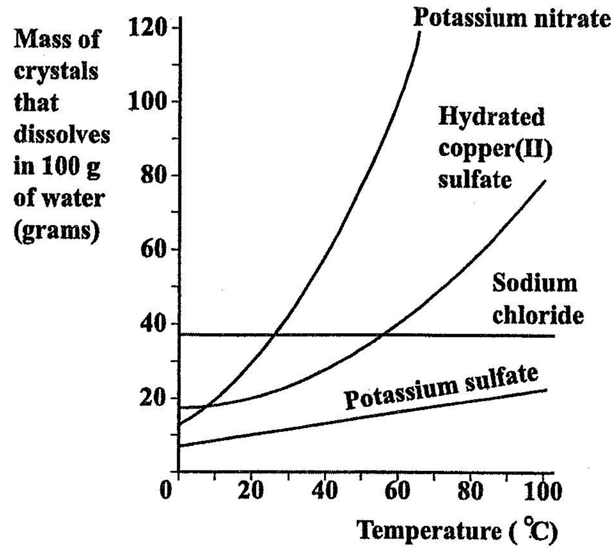
2. In the notation $^{35}_{17}Cl$, the subscript 17 represents the

(A) atomic number
(B) oxidation state
(C) mass of a mole of atoms
(D) atoms in a mole of molecules

3. From the table below, which of the substances represented by the options, A, B, C and D, is MOST likely sodium chloride?

Substance	Boiling Point (°C)	Electrical Conductivity	
		Solid State	Aqueous State
(A)	2600	Yes	No
(B)	444	No	No
(C)	1465	No	Yes
(D)	–35	No	Yes

Item 4 refers to the solubility curves in the following diagram.



4. The salt whose solubility changes to the GREATEST extent in the temperature range 0 °C to 60 °C is

(A) sodium chloride
(B) potassium nitrate
(C) potassium sulfate
(D) hydrated copper(II) sulfate

5. Which of the following substances supplies protons as the only positive ions in aqueous solutions?

(A) Salt
(B) Base
(C) Acid
(D) Alkali

6. Crystals of sodium chloride are BEST described as

(A) ionic
(B) metallic
(C) molecular
(D) macromolecular

7. The ionic equation for the reaction between an acid and a carbonate is represented by

(A) $2H^+(aq) + CO_3^{2-}(aq) \rightarrow HCO_3^-(aq)$

(B) $H^+(aq) + CO_3^{2-}(aq) \rightarrow H_2CO_3(aq)$

(C) $2H^+(aq) + CO_3^{2-}(aq) \rightarrow CO_2(g) + H_2O(l)$

(D) $H^+(aq) + CO_3^{2-}(aq) \rightarrow CO_2(g) + H_2O(l)$

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8. When 100 cm^3 of 2 mol dm^{-3} hydrochloric acid (in excess) are added to 5.0 g of granulated zinc, hydrogen gas is evolved at a certain rate. If 5.0 g of powdered zinc were used instead, the rate of production of hydrogen gas would

(A) increase because of the greater concentration of powdered zinc
(B) increase because of the increased surface area of powdered zinc
(C) decrease because of the decreased surface area of powdered zinc
(D) decrease because of the lower solubility of powdered zinc

9. An element, X, has an electronic configuration 2, 8, 1. At which of the following positions is the element in the periodic table?

(A) Period 1 Group I
(B) Period 1 Group II
(C) Period 3 Group I
(D) Period 3 Group II

Items 10–11 refer to the four sets of properties represented by the options A, B, C and D below.

	Relative Charge	**Approximate Mass**
(A)	+1	1
(B)	0	1
(C)	–1	0
(D)	0	2

In answering Items 10–11, each option may be used once, more than once, or not at all.

10. Which of the properties above refer to a neutron?

11. Which of the properties above refer to a proton?

12. Two solutions are to be mixed in order to demonstrate an endothermic change. Which of the following techniques would be MOST appropriate?

(A) Taking mass readings
(B) Taking temperature readings
(C) Monitoring the pH of the solutions
(D) Carefully observing colour changes

13. Sulfur and oxygen are in the same group of the periodic table because

(A) they can react with each other
(B) they can form covalent compounds
(C) they have the same number of electrons in their outer shell
(D) the atomic number of sulfur is 16 and the relative atomic mass of oxygen is 16

Item 14 refers to the reaction represented by the equation below.

$$CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$$

ethanoic acid + ethanol → ester + water

14. How many moles of ethanoic acid are required to produce 0.5 mol of the ester?

(A) 0.05
(B) 0.5
(C) 1
(D) 2

15. Which of the following compounds is covalent?

(A) Potassium chloride
(B) Calcium chloride
(C) Hydrogen chloride
(D) Magnesium chloride

Items 16–17 refer to the following acids.

(A) Ethanoic acid
(B) Hydrochloric acid
(C) Nitric acid
(D) Sulfuric acid

In answering Items 16–17, each option may be used once, more than once, or not at all.

16. Which of the acids above has a basicity of 2?

17. Which of the acids above is weakly ionized in aqueous solutions?

18. Which of the following reacts with an acid, liberating a gas which turns lime water milky?

(A) Methyl orange
(B) Barium chloride
(C) Calcium carbonate
(D) Magnesium metal

19. Which of the following gases shows NO reaction with moist litmus paper?

(A) Ammonia
(B) Hydrogen
(C) Sulfur dioxide
(D) Hydrogen sulfide

20. Which of the following properties increases across Period 3 of the periodic table?

(A) Atomic number
(B) Atomic radius
(C) Tensile strength
(D) Metallic character

Item 21 refers to the following information.

A solution of iron(II) sulfate was added until in EXCESS to a solution of barium nitrate. The precipitate produced was filtered off and water was added to the residue in the filter paper.

21. The colour of the precipitate produced was

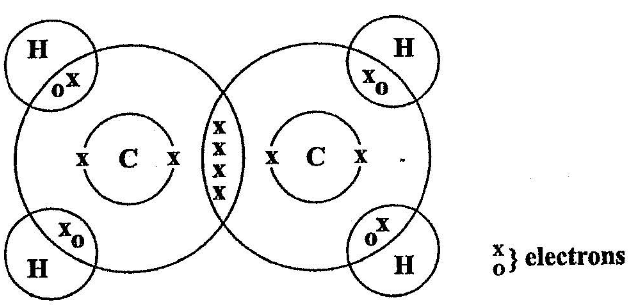
(A) yellow
(B) brown
(C) green
(D) white

22. What mass of oxygen atoms contains the same number of moles as 112 g of sulfur atoms?

[RAM: O = 16; S = 32]

(A) 0.56 g
(B) 5.60 g
(C) 56.00 g
(D) 560.00 g

Item 23 refers to the following diagram.



23. The bond between the two carbon atoms is

(A) single
(B) double
(C) triple
(D) dative

24. Which of the following halogens is a liquid at room temperature?

(A) Iodine
(B) Fluorine
(C) Bromine
(D) Chlorine

25. A catalyst increases the rate of a reaction because it

(A) binds irreversibly with the reactants
(B) binds reversibly with the reactants
(C) provides an alternative route of higher activation energy
(D) provides an alternative route of lower activation energy

26. Which of the following reactions occurs between propene and bromine?

(A) Addition
(B) Substitution
(C) Precipitation
(D) Condensation

27. $FeCl_2$ and $FeCl_3$ are two chlorides of iron. Which of the following statements about these two chlorides of iron are true?

I. The percentage of iron by mass in the two chlorides is different.
II. The colours of the aqueous solutions of the two chlorides are different.
III. The aqueous solutions of the two chlorides do not conduct electricity.

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

28. When solid lead nitrate is heated, it decomposes giving off nitrogen(IV) oxide and oxygen. The BALANCED equation for this reaction is

(A) $2Pb(NO_3)_2(s) \rightarrow 2PbO(s) + 4NO_2(g) + O_2(g)$

(B) $Pb(NO_3)_2(s) \rightarrow PbO(s) + NO_2(g) + O_2(g)$

(C) $Pb(NO_3)_2(s) \rightarrow PbO(s) + 2NO_2(g) + O_2(g)$

(D) $2Pb(NO_3)_2(s) \rightarrow PbO(s) + 2NO_2(g) + O_2(g)$

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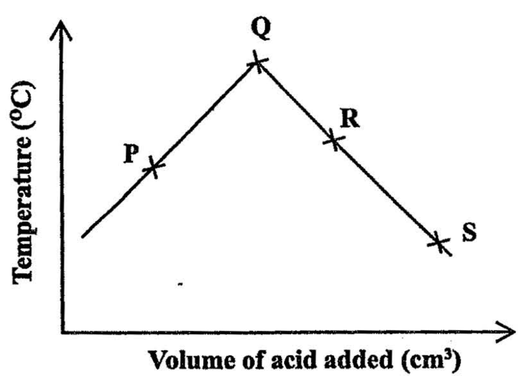
29. When a metal atom becomes an ion, it

(A) gains electrons and is oxidized
(B) gains electrons and is reduced
(C) loses electrons and is reduced
(D) loses electrons and is oxidized

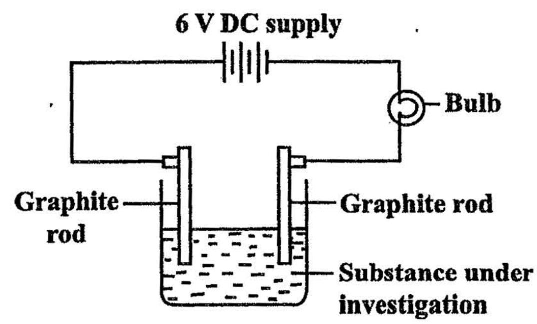
30. A solution of sodium hydroxide is neutralized by the addition of dilute hydrochloric acid. The results obtained are used to plot the graph below.



Which of the points on the graph above represents the neutralization point of the reaction?

(A) P
(B) Q
(C) R
(D) S

Item 31 refers to the following apparatus which is used to measure the relative conductivity of various substances.

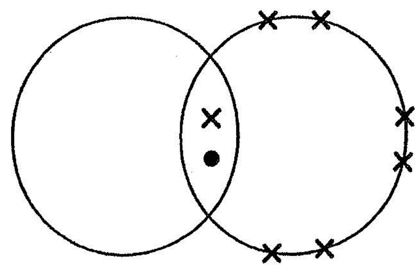


31. If substances containing one mole of solute per dm^3 are investigated, which substance should cause the bulb to glow BRIGHTEST?

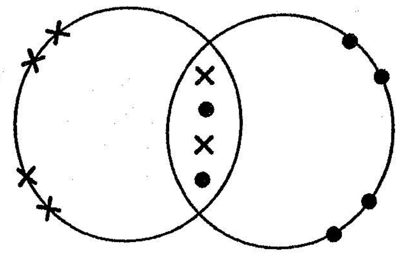
(A) Water
(B) Ammonia
(C) Sulfuric acid
(D) Ethanoic acid

<u>Items 32–33</u> refer to the following diagrams which represent bonding as it occurs in certain compounds.

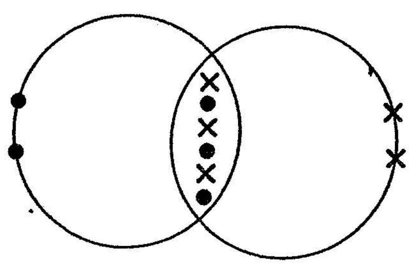
(A)



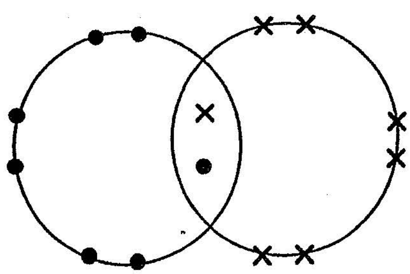
(B)



(C)



(D)



In answering Items 32–33, each option may be used once, more than once or not at all.

32. Which of the diagrams illustrates bonding in chlorine?

33. Which of the diagrams illustrates bonding in oxygen?

Item 34 refers to the reaction represented by the equation below.

$RCOOH(aq) + NaOH(aq) \rightarrow RCOONa(aq) + H_2O(l)$

34. Which of the following processes correctly describes the reaction above?

(A) Decomposition
(B) Neutralization
(C) Precipitation
(D) Reduction

Item 35 refers to the following information.

A compound when treated with concentrated hydrochloric acid gives off a greenish-yellow gas which bleaches moist red litmus paper.

35. The gas given off is

(A) chlorine
(B) sulfur dioxide
(C) hydrogen sulfide
(D) nitrogen dioxide

36. Which reaction involves the boiling of a fat or oil with aqueous sodium hydroxide (caustic soda)?

(A) Hydrolysis
(B) Dehydration
(C) Saponification
(D) Esterification

37. A current of 5 amperes was passed for 193 seconds through a solution of copper(II) sulfate using copper electrodes.
[RAM: Cu = 65; Faraday's constant, F = 96 500 C mol^{-1}]

The mass of copper that would be gained at the

(A) anode is 0.005 g
(B) cathode is 0.005 g
(C) anode is 0.325 g
(D) cathode is 0.325 g

38. The compound ethene is described as being unsaturated. This means that the

(A) carbon atoms in ethene are linked by single bonds
(B) carbon atoms in the molecule are very reactive
(C) molecule contains at least one double bond
(D) molecule has insufficient hydrogen atoms

39. Which of the following metals will react MOST vigorously with dilute acid?

(A) Copper
(B) Iron
(C) Lead
(D) Zinc

40. Which of the following methods is used for the extraction of aluminium?

(A) Electrolysis of its molten oxide
(B) Reduction of its oxide using coke
(C) Electrolysis of its aqueous chloride
(D) Reduction of its oxide using carbon monoxide

41. Which of the following metals is an important constituent of chlorophyll?

(A) Iron
(B) Copper
(C) Calcium
(D) Magnesium

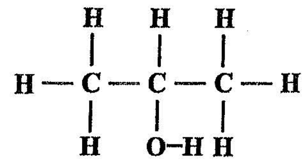
42. Which of the following options lists ALL the raw materials used in the extraction of iron?

(A) Air, iron ore and limestone
(B) Air, coke, iron ore and limestone
(C) Air, coke, cryolite and limestone
(D) Coke, iron ore, bauxite and limestone

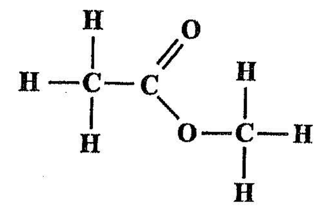
43. Which of the following compounds may be used to counteract the effects of acid rain on the soil?

(A) H_2S
(B) CaO
(C) CO
(D) SO_2

<u>Item 44</u> refers to the following compounds.



P

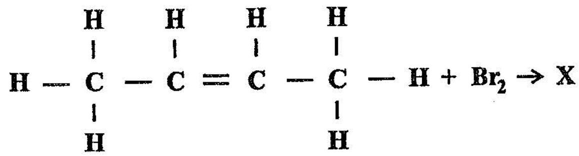


Q

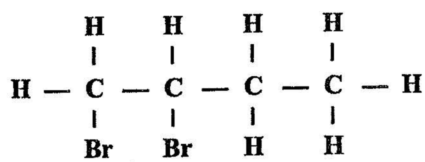
44. Which of the following MOST likely identifies P and Q?

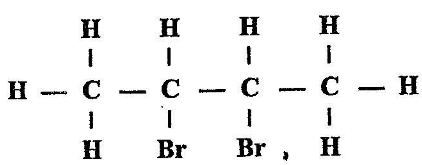
	P	Q
(A)	Acid	Alcohol
(B)	Ester	Acid
(C)	Alcohol	Acid
(D)	Alcohol	Ester

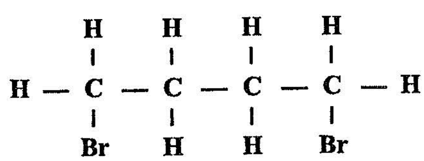
Item 45 refers to the following equation.

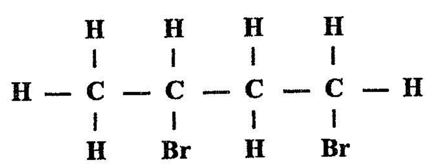


45. What is the correct structural formula for X?

(A) 

(B) 

(C) 

(D) 

Items 46–47 refer to the following compounds.

(A) CH_3COOH
(B) C_2H_5OH
(C) CH_3COOCH_3
(D) CH_3COONa

In answering Items 46–47, each option may be used once, more than once, or not at all.

46. Which of the compounds above has a fruity odour?

47. Which of the compounds above reacts with sodium carbonate to produce a gas which turns lime water milky?

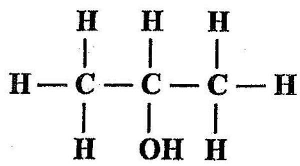
48. Which of the following is produced when carbon monoxide reacts with iron(III) oxide?

(A) O_2 only
(B) Fe only
(C) CO_2 and CO only
(D) CO_2 and Fe only

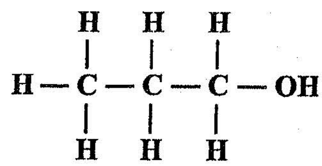
49. Hexane is an alkane with six carbon atoms per molecule. Which of the following is the formula for hexane?

(A) C_6H_{12}
(B) C_6H_{14}
(C) C_6H_{16}
(D) C_6H_{18}

Item 50 refers to two organic compounds, I and II, of molecular formula C_3H_8O, with the following structures:



Compound I



Compound II

50. Compounds I and II are known as the

(A) isomers of C_3H_8O
(B) isotopes of C_3H_8O
(C) condensed formulae of C_3H_8O
(D) molecular formulae of C_3H_8O

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51. Steel is often used in place of iron because it

(A) is more susceptible to corrosion
(B) has a lower melting point
(C) is stronger
(D) is cheaper

52. Which of the following statements is/are true of alkenes?

I. They react with $KMnO_4/H^+$.
II. They undergo addition reactions.
III. They undergo substitution reactions.

(A) I only
(B) I and II only
(C) I and III only
(D) I, II and III

53. Which of the following tests can be used to distinguish between an alkane and an alkene?

I. Adding bromine solution
II. Burning
III. Conductivity

(A) I only
(B) II only
(C) III only
(D) I and III only

54. What type of reaction occurs when zinc metal is added to copper sulfate solution?

(A) Combustion
(B) Neutralization
(C) Displacement
(D) Decomposition

55. During the manufacture of ethanol by fermentation, the gas evolved

(A) turns lime water cloudy
(B) burns with a blue flame
(C) relights a glowing splint
(D) turns potassium dichromate green

56. Which of the following is a natural source of hydrocarbons?

(A) Carbon
(B) Methane
(C) Hydrogen
(D) Petroleum

57. ALL members of a homologous series have similar

(A) densities
(B) boiling points
(C) chemical properties
(D) physical properties

58. Which of the following is the reason for adding cryolite to pure alumina in the extraction of aluminium by electrolysis?

(A) It lowers the melting point.
(B) It reduces the resistance.
(C) It reduces the current.
(D) It prevents damage.

59. Which of the following substances is a product of a halogenation reaction with methane?

(A) Nitrogen(IV) oxide
(B) Hydrogen chloride
(C) Ammonia
(D) Water

60. Which of the following compounds is NOT formed from condensation polymerization?

(A) Polysaccharides
(B) Polyalkenes
(C) Polyamides
(D) Polyesters

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.